

ATP3 and FX4021 came closer to doing it than the rest did. The results were very similar in *Behind Enemy Lines*, but this time, the explosions were better reproduced by the Logitech X-230, while the Altec Lansing duo did not do badly at all. The engine roar and the insane treble in *The Fast and the Furious* clip was reproduced better by the aforementioned trio. Again, as in the case of music tests, the Artis S444 and S800 did a great job, but were marginally behind the Logitech X-230 and the Altec Lansing ATP3 and FX4021.

DVD Audio

The Altec Lansing ATP3 was stronger in case of DVD Audio, producing all the fine details in the treble, along with strong vocals as well as bass. The Altec Lansing FX4021, Logitech X-230, and Artis S444 and S800 were just a few steps behind.

Game Performance

The Altec Lansing ATP3 kept us engrossed in the gaming test, and by far proved to be the best gaming 2.1 set. The Logitech X-230 was the next-

best, and was beaten because of the former's superior ability to better reproduce the different sounds and voices—even amidst the loud explosions. The FX4021 had overpowering bass, and as a result, lost quite a bit of the detail.

Frequency tests

The 50 Hz hum is a very difficult sound to produce for most speaker sets, and only the Altec Lansing ATP3 and Logitech X-230 produced respectable results. Overall, the Logitech was better in this test than the other sets, but some others—such as the Altec Lansing FX4021 as well as the Zebronics ZEB-SW8000—also performed well. The ATP3, FX4021, and the Z-230 passed the *ultimate bass* and *ultimate treble* tests very well indeed, though none could clear the *ultimate bass* test at a volume above 75 per cent.

MY HOME, MY THEATRE: 5.1 SPEAKERS

For those of us who don't like to be confined by the shackles of stereo sound, 5.1-channel speaker sets are the answer. Almost all DVD movies and games today boast of surround sound and 5.1-channel audio. To meet these aural requirements, all modern motherboards come with a minimum of six-channel audio. Even HD audio is becoming the norm. 5.1-channel systems enrich the listening experience and create a more immersive audio environment.

Unlike their 2.1 counterparts, 5.1 channel speaker sets derive sound from six discrete sound channels. One channel goes to the centre speaker, which is responsible for the main functions such as dialogue delivery. Two go to the rear left and right, and two to the front left and right. These are responsible for creating a 3D map of the audio environment around the listener. The last channel goes to the subwoofer, which is responsible for delivering low frequency sounds such as explosions and deep drum beats.

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New Age A couple of Enigma tracks were chosen in this genre. Songs of this genre have some very good and reverberating beats with tight bass.

Acid Trance We looked for clear and faithful reproduction of the tight beats and special effects, and also noted the audibility of the various instruments in a couple of tracks from *The Vicious Spiral*. Treble was also noted here.

Western Classical In this genre, we had compositions with a myriad of instruments and covering the widest audio spectrum—a string quintet by Mozart and a symphony by Dvorak. We noted the audibility at high and low volumes, the treble, especially the piano and the violin and the quality of bass.

Our Indian playlist consisted of:

Bollywood *Tu Hi Meri Shabh* from the movie *Gangster*. The clarity, distortion of vocals during heavy drumming, and distortion in the chorus were noted.

Bhangra *Tunak Tunak Tun* by Daler Mehndi. In this song, we paid attention to the drum beats and the *Ektara*.

Ghazal *Hoshwalon Ko Khabar Kya* by Jagjit Singh. This song has reverberating vocals and instrumental sounds, and we paid attention to these aspects.

Indian Classical Vocal *Mundana* by Pt. Bhimsen Joshi. In this

song, the voice of the singer traverses from low to high in a very little space of time, and we looked for distortions when the speaker sets attempted to handle his powerful voice.

Indian Classical Instrumental This track sample had the *Santoor* and *Tabla*. We carefully evaluated whether the speaker sets could reproduce individual beats on the *Santoor* strings and the sound of fingers striking the surface of the *Tabla*. We also listened to the quality of the *Tabla* beats. We also used special test files of a constant frequency to test low-, mid- and high-frequency across the audio spectrum. These ranged from a humming 50 Hz to a shrill, ear-piercing 15 kHz. The treble and bass of all the speaker sets was tested by playing special THX-certified "Air Raid" and "Ultimate Bass Test" sound samples from Creative.

WinAMP 5.3 was used to play all the sound files, and no extras such as DSP or visualisation plug-ins were installed. The equalisers were set to Off. For the DVD, we used CyberLink PowerDVD 7.2 Ultra BD Edition. The tests were carried out at varying volume levels, and in the case of separate bass and treble controls, these were kept at 50 per cent through the first run and 100 per cent through the second run.

We also noted the different features (or lack of them) for the different speaker sets such as the various inputs, connectivity, controls, accessories, cables, and build quality.